

Probabilistic Soft Logic (PSL)

AIEA Lab Onboarding Task 5

What is PSL

- A machine learning + logic framework
- Combines logical rules with probability
- Uses values from 0 to 1 (soft truth) instead of just true/false
- Designed for reasoning with uncertainty

Why should we use PSL?

- Real-world data is often uncertain or incomplete
- PSL allows:
 - Partial relationships
 - Confidence levels
- Used in:
 - Social networks
 - Recommendation systems
 - Link prediction

How does PSL work?

- Define rules (.psl file)
- Provide data (.data file)
- Run inference
- Get predictions (0–1 values)
- Uses optimization (ADMM) to find best results

Example Rule

20: Lived(P1, L) & Lived(P2, L) $\rightarrow$ Knows(P1, P2) 2

- Same location $\rightarrow$ more likely to know each other
- 20 = importance (weight)
- 2 = smoother learning

Running PSL + Output

Example Inputs:

```
git clone https://github.com/linqs/psl-examples.git
```

```
cd simple-acquaintances/cli
```

```
./run.sh
```

Example Outputs:

```
'Alex' 'Arti' 0.9966
```

```
'Alex' 'Ben' 0.5923
```

- Closer to 1 = stronger relationship

Conclusion

- PSL = logic + probability
- Handles uncertainty better than traditional logic
- Easy to run with examples
- Useful for real-world AI problems